MILITARY SPECIFICATION SHEET

RELAY, VACUUM, SPST, N/C, 15 AMPERES DC OR
60 HZ RMS, 12 KILOVOLTS DC OR 60 HZ PEAK

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-R-83725.

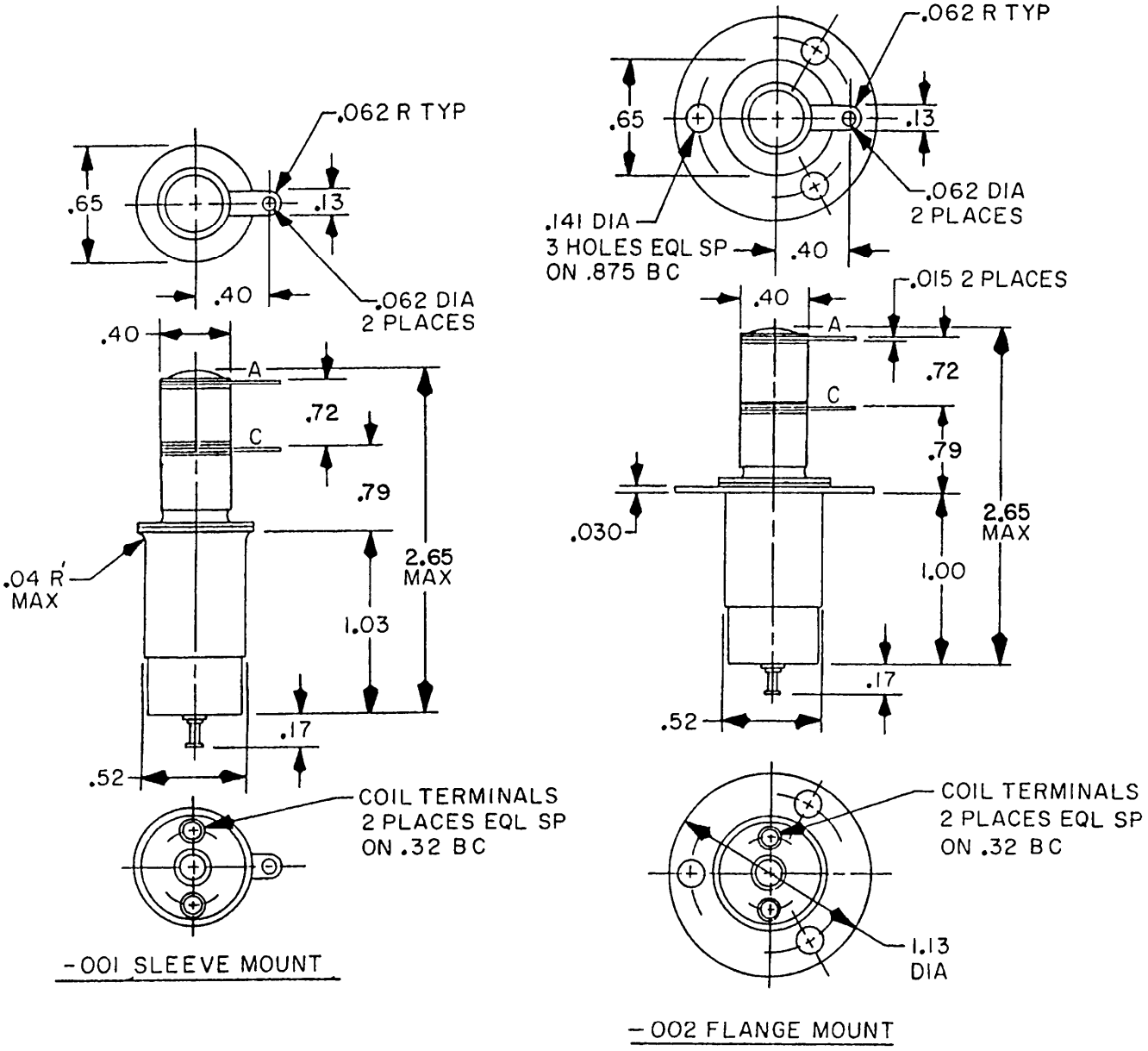
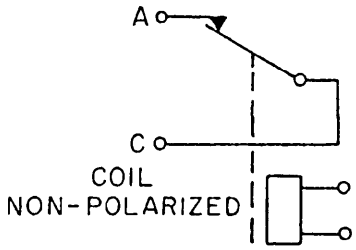


FIGURE 1. Relay, dimensions and configurations.

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CIRCUIT DIAGRAM

INCHES	mm	INCHES	mm
.015	.38	.52	13.2
.030	.76	.65	16.5
.040	1.02	.72	18.3
.062	1.57	.79	20.1
.13	3.3	.875	22.2
.141	3.58	1.00	25.4
.17	4.3	1.03	26.2
.32	8.1	1.13	28.7
.40	10.2	2.65	67.3

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .010 (0.25 mm).
3. Metric equivalents are given for general information only.
4. Shape of coil lug (solder) terminals optional, however they must accomodate two 22 AWG wires.
5. Ceramic insulator is not glazed.
6. M83725/19-001 is mounted by .52 inch (13.2 mm) diameter of body of coil housing or by .65 inch (16.5 mm) diameter lip in center of relay.

FIGURE 1. Relay, dimensions and configurations - Continued.

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REQUIREMENTS:

Contact data:

Configuration: SPST, N/C, ground isolated.

Arrangement: 1 form B.

Load ratings:

Resistive: (Carry only).
15 amperes dc or 60 Hz rms.
12 amperes rms, 2.5 MHz.
6 amperes rms, 16 MHz.
4 amperes rms, 32 MHz.

Rated operating voltage:
12 kilovolts peak, 60 Hz or dc.
12 kilovolts peak, 2.5 MHz.
9 kilovolts peak, 16 MHz.
7 kilovolts peak, 32 MHz.

Contact resistance:

Rated life:
Before: .020 ohm, maximum.
During: .100 ohm, maximum.
After: .100 ohm, maximum.

Contact bounce: Not applicable.

Capacitance:
0.8 picofarad, maximum across open contacts.
0.8 picofarad, maximum between open contacts and ground.

Coil data:

Duty rating: Continuous.

Maximum voltage: 32 V dc.

Nominal voltage: 26.5 V dc.

Pickup voltage: 22 V dc, maximum over temperature range.

Dropout voltage: 1 V dc, minimum over temperature range.

Coil resistance: 290 ohms ±10 percent.

Operate time: 15 milliseconds, maximum over temperature range.

Release time: 15 milliseconds, maximum over temperature range.

Electrical data:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and housing at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

At atmospheric pressure: 60 Hz.
Between all mated contacts in the open position: 17.5 kV peak.
Between high voltage terminals and housing: 17.5 kV peak.
Between coil and housing: 500 V rms, 60 Hz.

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In a suitable insulating medium such as a fluorocarbon insulating liquid or tightly adherent and void free encapsulant: 60 Hz.
Between all mated contacts in the open position: 24 kV peak.
Between high voltage terminals and housing: 24 kV peak.
Between coil and housing: 500 V rms, 60 Hz.

Ground isolated: Ground isolated relays have operating elements insulated from the ground plane to provide absolute voltage isolation between the frame and the high voltage contacts.

Environmental data:

Temperature range: -55°C to +125°C.

Vibration: MIL-STD-202, method 204, test condition A (10-55 Hz, .06 inch DA/55-500 Hz, 10 G).

Shock: MIL-STD-202, method 213, test condition J, 30 G, 11 milliseconds, half sine.

Physical data:

Terminal strength: 5 pounds pull.

Dimensions and configuration: See figure 1.

Termination: Solder terminal.

Weight: 1.1 ounces, maximum.

Life test requirements:

Mechanical cycling: 1,000,000 cycles. 2 sample units (cycling rate, 36,000 per hour maximum; contact current shall not exceed 10 milliamperes).

Quality assurance:

Dielectric withstanding voltage:

Tests to be conducted at atmospheric pressure rating only.
Duration of application: 5-10 seconds at a 10 percent increase in the dielectric withstanding voltage.

Part number:
M83725/19-0001 sleeve mount.
M83725/19-0002 flange mount.

Custodian:
Air Force - 85

Review activity:
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F615)

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